

Shaw Sun . Jarvis Consulting

I recently graduated from the University of Guelph with a Master of Data Science and have experience in research and industry settings working on data pipelines, data quality, analytics, and machine learning projects. My work has involved Python, SQL, Excel, ETL development, workflow monitoring, troubleshooting, and model evaluation, with a focus on building reliable and practical data solutions. I am seeking Data Analyst/Engineer/Scientist, or Machine Learning-related roles where I can apply both technical and problem-solving skills. I offer a strong foundation in data work, clear communication, careful analysis, and the ability to support projects from data preparation to evaluation and reporting.

Skills

Proficient: Python(Pandas/Numpy), SQL, Bash, ETL pipelines, Git, Agile/Scrum

Competent: Scikit-Learn, Databricks, LLM Fine-Tuning, Machine Learning Models, R/R studio/Rshiny

Familiar: PySpark, PyTorch, Java, C#, Excel/Pivot Table

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_ShawSun

Cluster Monitor [GitHub]: Developed a Linux Cluster Monitoring Agent for the Jarvis LCA team to support resource planning in a Rocky Linux cluster. The agent uses Bash scripts to collect both static host information, such as hostname, CPU, memory, and architecture, and real-time usage metrics, such as CPU and memory usage. The data is inserted into PostgreSQL, allowing users to run SQL queries for usage analysis and capacity planning. PostgreSQL was provisioned with Docker, while cron scheduled the usage script to run automatically at regular intervals. I manually tested the scripts, verified database inserts, and managed the source code with Git and GitHub.

Python Data Analytics [GitHub]: Developed a Python Data Analytics project to analyze an online retail dataset from London Gift Shop and extract business insights from customer transactions. The project loads retail data from PostgreSQL into a pandas DataFrame, then performs data wrangling steps such as handling missing values, creating invoice-level metrics, calculating monthly revenue, identifying placed and canceled orders, and separating new versus existing customers. I used pandas and NumPy for data transformation and Matplotlib to visualize trends such as monthly sales, order patterns, and customer behavior. The analysis results help understand revenue changes, customer growth, and purchasing patterns over time.

Highlighted Projects

Stock Prices Prediction via LLM: Implemented a stock price prediction project using Reuters news data to support research and backtesting, implemented a scalable ETL pipeline for 10M+ new records and standardized JSON-to-tabular ingestion while operationalizing Llama2-13B embeddings as sentiment and feature inputs for downstream ML models, evaluated performance with log loss, accuracy, and return-based metrics, and improved predictive stability and estimated strategy ROI through feature and threshold tuning.

Text2SQL system with RAG [GitHub]: Developed a RAG-based Text2SQL system for complex query generation, using DeepSeek-R1:8B, SentenceTransformers, and FAISS to retrieve relevant schema context and improve prompt grounding for multi-table and nested SQL tasks, tested the system on the Spider benchmark and analyzed error patterns with Ridge Regression bias-variance decomposition, resulting in an accuracy gain from 16% to 72% and clearer insight into underfitting on smaller schemas.

Phishing Email Detector [GitHub]: Built a phishing email detection system for multi-domain email datasets, implemented a fine-tuned BERT model with stratified sampling, distributed training, and dynamic learning-rate scheduling to improve training efficiency and maintain robustness, tested performance across datasets including SpamAssassin, Enron, and Kaggle, and delivered a Streamlit dashboard for risk and compliance teams to inspect model decisions, monitor false positives, and simulate email threat scenarios in real time.

Professional Experiences

Data Engineer, Jarvis Consulting Group (Apr 2026 - Present): Worked as a Data Engineer at Jarvis where I built a cluster monitoring project to collect server hardware and usage data for resource planning, and completed a retail analytics project that loaded transaction data from PostgreSQL, performed data wrangling, analyzed revenue and

order patterns, and visualized customer behavior trends, while continuing to contribute to additional ongoing projects in a fast-paced team environment.

Research Assistant (Prof. Zhenzhen Fan), University of Guelph (Nov 2025 - Jan 2026): Processed large-scale news datasets using Python and SQL to support market-narrative analytics research, cleaned and transformed raw news records into analysis-ready tables, and prepared standardized KPI datasets for downstream reporting and modeling tasks. Assisted with exploratory data analysis, checked data consistency during processing, summarized key patterns from news-related variables, and supported research interpretation through organized data outputs and clear documentation.

Research Assistant (Prof. Fred Liu), University of Guelph (May 2025 - Aug 2025): Analyzed financial and business datasets using Python, SQL, and Excel to support downstream reporting and financial modeling tasks, cleaned and transformed raw data for analysis, explored variable relationships and data patterns, and evaluated model results with statistical and machine learning metrics such as AUC, MAE, RMSE, and Sharpe ratio to support research findings and improve result interpretation.

Data Scientist (Intern), Baidu, Inc (May 2023 - Aug 2023): Supported an enterprise LLM product as a Data Scientist Intern at Baidu by preparing and transforming product-related datasets using SQL and Python, reviewing data quality issues such as duplicate records and inconsistent labels, and assisting with analysis and reporting tasks for model evaluation. Collaborated with team members to understand business requirements, troubleshoot data issues, and summarize data patterns to support product improvement.

Education

University of Guelph (Sep 2024 - Aug 2025), Master of Data Science, Department of Mathematics and Statistics - GPA: 3.9/4.0

Virginia Tech (Aug 2019 - Dec 2023), Bachelor of Science in Computational Modeling and Data Analysis, College of Science - Dean's List: Fall 2020, Spring 2021, Spring 2022

Miscellaneous

- Basketball playe: Represented Virginia Tech in NECBL on multiple occasions
- Long driving trip: A drive of less than five hours is not a long journey for me
- Game enthusiasts: I have played so many different kinds of games other than horror game. I even don't watch horror movie